SOLUTIONS

EE4-01
EE9-SC2

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020-2021

MSc and EEE/EIE PART IV: MEng and ACGI

ADVANCED COMMUNICATION THEORY
(Blackboard Remote Examination)

Tuesday 4th May 2021, 10:00 am (UK time)

Time allowed: 3:00 hours.

There are 20 questions on this paper.

Answer ALL questions.

All questions carry equal marks

Examiners responsible:

First Marker(s): Prof A. Manikas

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1 Path Loss

If the path-loss is 23.0103 dB and the power of the received signal is 6.9897 dBm, then the power of the transmitter is

(a) 20 W;
(b) 10 W;
(c) 1 W; [x]
(d) 10 mW;
(e) 20 mW.

2 Array Capacity for Channels with inf BW

Consider an array receiver employing a ULA (uniform linear array) of N antennas with half-wavelength inter-antenna spacing. If the bandwidth of the systems B tends to infinity and the capacity is

$$C = 10.08 P_s/(N_0 + N_j) \text{ bits/s}$$

where P_s is the desired signal power and the power spectral density of the noise $\text{PSD}_n(f)$ and of the co-channel interference/jammer $\text{PSD}_j(f)$ are

$$\text{PSD}_n(f) = \tfrac{N_0}{2}; \quad \text{PSD}_j(f) = \tfrac{N_j}{2}$$

the number of antennas N is

(a) $N = 4$;
(b) $N = 5$;
(c) $N = 6$;
(d) $N = 7$;[x]
(e) none of the above.

3 Rx Array: Covariance Matrix 1

Consider an antenna array system of 10 elements operating in the presence of one desired and two uncorrelated co-channel interfering signals, all of power $P_s = 3$. The power of the noise is $\sigma_n^2 = 10$. If $\mathbb{R}_{xx}$ is the theoretical covariance matrix of the received signal vector $\underline{x}(t)$, then which of the following statements is correct?

(a) The minimum eigenvalue of $\mathbb{R}_{xx}$ is equal to 3.
(b) The principal eigenvalue of $\mathbb{R}_{xx}$ is equal to 10.
(c) The rank of $\mathbb{R}_{xx}$ is equal to 3.
(d) The rank of $\mathbb{R}_{xx}$ is equal to 10.. [x]
(e) None of the above.

4 Rx Array: Covariance Matrix 2

Consider an antenna array system of 5 elements operating in the presence of one desired and two co-channel interfering signals all of power equal to 10. The power of the noise is 30 dB below the signal power. If $\mathbb{R}_{xx}$ is the theoretical covariance matrix of the received signal vector $x(t)$ then which of the following statements is correct?

(a) The rank of $\mathbb{R}_{xx}$ is equal to 2.

(b) The rank of $\mathbb{R}_{xx}$ is equal to 3.

(c) The minimum eigenvalue of $\mathbb{R}_{xx}$ is equal to 10.

(d) The minimum eigenvalue of $\mathbb{R}_{xx}$ is equal to 0.01. [x]

(e) None of the above.

5 Array Signal Subspace: Coherent Signals

Consider a single transmitter whose signal arrives at the receiver via 4 multipaths where the first two paths are partially correlated and other two are coherent (fully correlated). The receiver consists of a uniform linear antenna array of 10 antennas with half-wavelength inter-antenna spacing. The dimensionality of the signal subspace is:

(a) 1.

(b) 2.

(c) 3. [x]

(d) 4.

(e) 10.

6 Rx Array Cov Matrix: Eigenvectors

Consider an antenna array system of N elements operating in the presence M co-channel sources ($M < N$). If $\underline{S}_i$ is the manifold vector associated with the ith source and $\mathbb{E}_s$ and $\mathbb{E}_n$ denote the matrices whose columns are the signal eigenvectors and the noise eigenvectors respectively of data covariance matrix $\mathbb{R}_{xx}$ then which of the following expressions is correct?

(a) $\mathbb{E}_n.\mathbb{E}_n^H.\underline{S}_i = \underline{S}_i$.
(b) $\mathbb{E}_s.\mathbb{E}_s^H.\underline{S}_i = \underline{0}$.
(c) $\mathbb{E}_n.\mathbb{E}_n^H.\underline{S}_i = \underline{0}$. [x]
(d) $\left(\mathbb{I}_N - \mathbb{E}_s.\mathbb{E}_s^H\right).\underline{S}_i = \underline{S}_i$.
(e) None of the above

7 Antenna Array Aperture

Consider an array of 4 antennas with Cartesian coordinates given by the following matrix

$$\begin{bmatrix} -2, & 2, & 2, & -2 \\ -0.5, & 0, & 0.5, & 1 \\ 0, & 0, & 0, & 0 \end{bmatrix} \text{ in units of half-wavelengths}$$

The array aperture (in half-wavelengths) is

(a) 4;
(b) 4.0311;
(c) 4.1231; [x]
(d) 4.272;
(e) none of the above.

Answer

(a) r1=[-2,-0.5,0]'; r2=[2,0,0]'; r3=[2,0.5,0]';r4=[-2, 1,0]'; n12=norm(r1-r2), n13=norm(r1-r3), n14=norm(r1-r4), n23=norm(r2-r3), n24=norm(r1-r4), n34=norm(r3-r4), max([n12,n13,n14,n23,n24,n34])=n13=4.1231

8 Array Reference Point

Consider a uniform linear array of 3 antennas. The carrier frequency is 2.4 GHz and the manifold vector for a signal with Direction-of-Arrival ($\theta = 30^0, \phi = 0^0$) is

$$[-0.9127 + j0.4086, 0.2089 + j0.9779, 1.0000 + j0.0000]^T$$

The origin of the Cartesian coordinates (array reference point) is the

(a) 1st antenna;
(b) 2nd antenna;
(c) 3rd antenna; [x]
(d) the centroid of the array geometry.
(e) none of the above

9 Array Manifold Vector

Consider that one of the paths from the transmitter of a CDMA user arrives at the reference point of an antenna array CDMA receiver from direction (azimuth, elevation)= $(60°, 0°)$. For this path, if the Cartesian coordinates of the antenna array elements are given by the columns of the following matrix

$$[\underline{r}_1, \underline{r}_2] = \begin{bmatrix} -0.5, & +0.5 \\ 0, & 0 \\ 0, & 0 \end{bmatrix} \text{ in units of half-wavelength,}$$

then manifold vector is

(a) $\underline{S}(\theta) = \left[\frac{\sqrt{2}}{2} + j\frac{\sqrt{2}}{2}, \quad \frac{\sqrt{2}}{2} - j\frac{\sqrt{2}}{2}\right]^T$; [x]

(b) $\underline{S}(\theta) = \left[-j\frac{\sqrt{2}}{2}, \quad +j\frac{\sqrt{2}}{2}\right]^T$;

(c) $\underline{S}(\theta) = \left[j, \quad -j\right]^T$;

(d) $\underline{S}(\theta) = \left[-j, \quad j\right]^T$;

(e) none of the above

10 Manifold: Rate of Change of Arclength

Consider a linear array of 5 Rx-antennas having the following Cartesian coordinates:

$$[\underline{r}_1, \underline{r}_2, \underline{r}_3, \underline{r}_4, \underline{r}_5] = \begin{bmatrix} -2, & -1, & 0, & +1, & +2 \\ 0, & 0, & 0, & 0, & 0, \\ 0, & 0, & 0, & 0, & 0, \end{bmatrix} \text{ in units of half-wavelength.}$$

The rate of change of the arclength $\dot{s}(\theta)$ of the array manifold for a source with Direction-of-Arrival (azimuth) $\theta = 30°$ is

(a) $\dot{s}(30°) = 19.631$;

(b) $\dot{s}(30°) = 9.9346$;

(c) $\dot{s}(30°) = 5.4414$;

(d) $\dot{s}(30°) = 4.9673$;[x]

(e) none of the above.

11 Beamformer: Steering Vector

Consider a beamformer which employes a uniform linear array of N antennas. The carrier frequency is 2.4 GHz and the manifold vector for a signal with Direction-of-Arrival ($\theta = 150^0, \phi = 0^0$) is

$$[-0.5902 + 0.8072i, 0.2089 - 0.9779i, 0.2089 + 0.9779i, -0.5902 - 0.8072i]^T$$

To steer the main lobe of the array towards the direction ($\theta = 150^0, \phi = 0^0$), the weight vector should be

(a) $\begin{bmatrix} 1, & 1, & 1, & 1 \end{bmatrix}^T$;

(b) $[-0.5902 - 0.8072i, 0.2089 + 0.9779i, 0.2089 - 0.9779i, -0.5902 + 0.8072i]^T$; [x]

(c) $[+0.5902 - 0.8072i, -0.2089 + 0.9779i, -0.2089 - 0.9779i, +0.5902 + 0.8072i]^T$;

(d) $[-0.5902 + 0.8072i, 0.2089 - 0.9779i, 0.2089 + 0.9779i, -0.5902 - 0.8072i]^T$;

(e) none of the above.

12 Beamformer: Beamwidth

For a uniform linear array of 100 sensors, operating at 30GHz (mmWave) frequency and half-wavelength inter-antenna spacing, the beamwidth is

(a) 1.1460°;

(b) 1.5608°;

(c) 2.2920°; [x]

(d) 3.1016°;

(e) none of the above.

13 Beamformer: Pout, SNRout

Consider a steering vector beamformer which employs a uniform linear array of $N = 5$ antennas that operates in the presence of a single desired signal with direction ($\theta = 90^0, \phi = 0^0$). The carrier frequency is 2.4 GHz and the array steers its main lobe towards the direction ($\theta = 90^0, \phi = 0^0$). Consider that the received desired signal power is equal to 1 while the channel noise is assumed to be additive white Gaussian noise of power 10 dB below the signal power. If P_{out} denotes the power of the desired signal at the output of the beamformer and SNR_{out} represents the output signal-to-noise ratio, which of the following statements is correct?

(a) P_{out}=5 and SNR_{out}=10.

(b) P_{out}=25 and SNR_{out}=10.

(c) P_{out}=5 and SNR_{out}=50.

(d) P_{out}=25 and SNR_{out}=50. [x]

(e) None of the above.

14 Beamformer: Output Noise Power

Consider a beamformer which employes a uniform linear array of N antennas and uses the following weight vector:

$$[-0.1125 + 0.9936i, 0.6661 + 0.7458i, 1.0000, 0.6661 - 0.7458i, -0.1125 - 0.9936i]^T.$$

If the channel noise is additive white Gaussian noise with power $\sigma_n^2 = 10^{-4}$ then the noise power at the beamformer's output is

(a) 0.00025;

(b) 0.0005; [x]

(c) 0.005;

(d) 0.025;

(e) none of the above.

15 SIMO Capacity

Consider a SIMO communication system operating in the presence of a desired signal with signal power 100mW and one interferer with double-sided PSD(f) of 10^{-2} W/Hz. The noise is assumed to be AWGN with double-sided PSD(f) of 10^{-2} W/Hz. The receiver employs a complete interference cancellation subspace beamformer (with weight vector having norm equal to 1) and utilises a ULA of 10 antennas. If the channel bandwidth is 16 kHz the system/channel capacity is

(a) 70.2001 bits/sec;

(b) 71.4003 kbits/sec;

(c) 72.0223 kbits/sec; [x]

(d) 143.8205 kbits/sec;

(e) ∞ bits/sec.

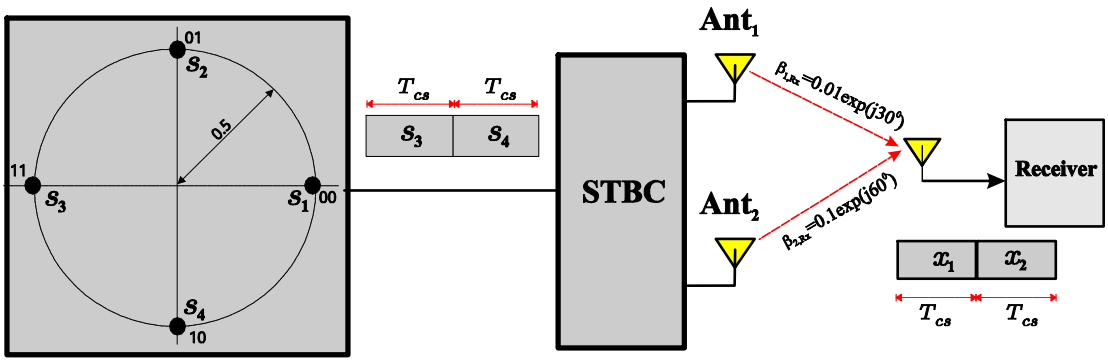
Answer

N=10; B=16000;N_0=2*10^{-2} P_s=0.1; P_n=B*N_0; C=B*$\log_2$(1+N*Ps/Pn)=16000*$\log_2$(1+ $10 * 0.1/(16000 \times 2 \times 10^{-2}) = 72.022$

16000*$\log_2(1 + 10 * 0.1/(16000 \times 10^{-2}) = 143.82$

16 MISO STBC

Consider the QPSK MISO system of 2 Tx antennas operating in a frequency flat wireless channel as shown the following figure:



If the QPSK symbols $[s_3, s_4]$ are transmitted using the above "Space-Time Block Coder" (STBC) then the receiver's input $[x_1, x_2]$, ignoring the noise, is

(a) $\Big[+0.0173 + j0.0458, \quad -0.0298 + j0.42\Big]$;

(b) $\Big[-0.0173 + j0.0458, \quad -0.0298 - j0.42\Big]$; [x]

(c) $\Big[-0.0173 - j0.0458, \quad -0.0298 + j0.42\Big]$;

(d) $\Big[+0.0173 - j0.0458, \quad -0.0298 - j0.42\Big]$;

(e) none of the above.

17 Spatiotemporal Manifold Vector

In a Space-Time DS-CDMA system employing a PN-code of length N_c (with a chip period T_c) and an array of N antennas at the receiver, which of the following statements involving the STAR manifold vector $\underline{h}(\theta, \ell)$ of a path is correct?

(a) $\|\underline{h}(\theta, \ell)\|^2 = NN_c$ [x]
(b) $\|\underline{h}(\theta, \ell)\|^2 = N\ell$
(c) $\|\underline{h}(\theta, \ell)\|^2 = N(N_c - \ell)$
(d) $\|\underline{h}(\theta, \ell)\|^2 = N_c\ell$
(e) None of the above

N.B.: The parameter $\ell \triangleq \left\lceil \frac{\tau}{T_c} \right\rfloor \mathrm{mod} N_c$ is the discrete version of the path delay τ. The parameter θ denote the azimuth angle.

18 MIMO: Virtual Array

With reference to a MIMO wireless communication system, consider that both the Tx and Rx antenna arrays are linear arrays with the Cartesian coordinates of their elements given by the columns of the following matrices

$$\mathrm{Tx} : [\underline{r}_1, \underline{r}_2] = \begin{bmatrix} 0, & 0 \\ -0.5, & 0.5 \\ 0, & 0 \end{bmatrix} \text{ in units of half-wavelength.}$$

$$\mathrm{Rx} : [\underline{r}_1, \underline{r}_2] = \begin{bmatrix} 0, & 0 \\ -2, & 2 \\ 0, & 0 \end{bmatrix} \text{ in units of half-wavelength.}$$

Which of the following statements, associated with its virtual Rx antenna array of an equivalent SIMO wireless communication system, is correct?

(a) $\begin{bmatrix} -2, & 2, & -2, & 2 \\ -0.5, & -0.5, & 0.5, & 0.5 \\ 0, & 0, & 0, & 0 \end{bmatrix}$ (i.e. a planar array).

(b) $\begin{bmatrix} -0.5, & -0.5, & 0.5, & 0.5 \\ -2, & 2, & -2, & 2 \\ 0, & 0, & 0, & 0 \end{bmatrix}$ (i.e. a planar array).

(c) $\begin{bmatrix} -2.5, & 1.5, & -1.5, & 2.5 \\ 0, & 0, & 0, & 0 \\ 0, & 0, & 0, & 0 \end{bmatrix}$ (i.e. a linear array).

(d) $\begin{bmatrix} 0, & 0, & 0, & 0 \\ -2.5, & 1.5, & -1.5, & 2.5 \\ 0, & 0, & 0, & 0 \end{bmatrix}$ (i.e. a linear array). [x]

(e) None of the above.

19 DoF

In a properly designed antenna array system the number of degrees-of-freedom (DoF) increases by

(a) decoupling "space" and "time" processes;

(b) increasing the array aperture;

(c) increasing the number of antennas N; [x]

(d) increasing the signal-to-noise ratio (SNR);

(e) none of the above.

20 Antenna Array: Wideband Assumption

If $\underline{x}(t)$ denotes the received signal vectors from a single transmitter, transmitting the baseband signal $m(t)$, and $\mathbb{R}_{xx} = \mathcal{E}\left\{\underline{x}\,(t)\,\underline{x}\,(t)^H\right\}$ the received data covariance matrix, which of the following equations is related to the "wideband" (WB) assumption?

(a) $\underline{x}(t) = \underline{S}m(t) + \underline{n}(t)$

(b) $\mathbb{R}_{xx} = P_m \underline{S}\underline{S}^H + \mathbb{R}_{nn}$

(c) $\mathbb{R}_{xx} = \underline{S}\underline{S}^H \mathbb{R}_{\underline{\mathrm{mm}}}$

(d) $\mathbb{R}_{xx} = \underline{S}\underline{S}^H \odot \mathbb{R}_{\underline{\mathrm{mm}}}$

(e) none of the above, [x]

where

$$\underline{S} = \underline{S}(\theta, \rho, \text{array geometry, carrier freq})$$

is the "spherical wave manifold vector", θ denotes the Direction-of-Arrival, ρ is the range and

$$\begin{aligned} \mathbb{R}_{\underline{\mathrm{mm}}} &= \mathcal{E}\left\{\underline{\mathrm{m}}\,(t)\,\underline{\mathrm{m}}\,(t)^H\right\} \\ \mathbb{R}_{nn} &= \mathcal{E}\left\{\underline{n}\,(t)\,\underline{n}\,(t)^H\right\} \end{aligned}$$

with $\underline{\mathrm{m}}\,(t)$ denoting the vector with elements the signal $m(t)$ arriving at each element of the antenna array.